

UPI Transaction Performance Diagnostics

An end-to-end performance database diagnostics evaluation mapping execution parameters, peak operational hour metrics, and merchant ecosystem insights.

LEAD ANALYST

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REPOSITORY CORE

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Executive Summary Dashboard

DIAGNOSTIC CORE OVERVIEW

Total Spend Volume

₹4,13,198.86

Aggregated across 150 unique transactions

The total transaction ledger registers highly stable volume velocity across base user structures. Digital transaction models continue to replace standard fiat channels cleanly across all tested segments.

System Operational Frictions

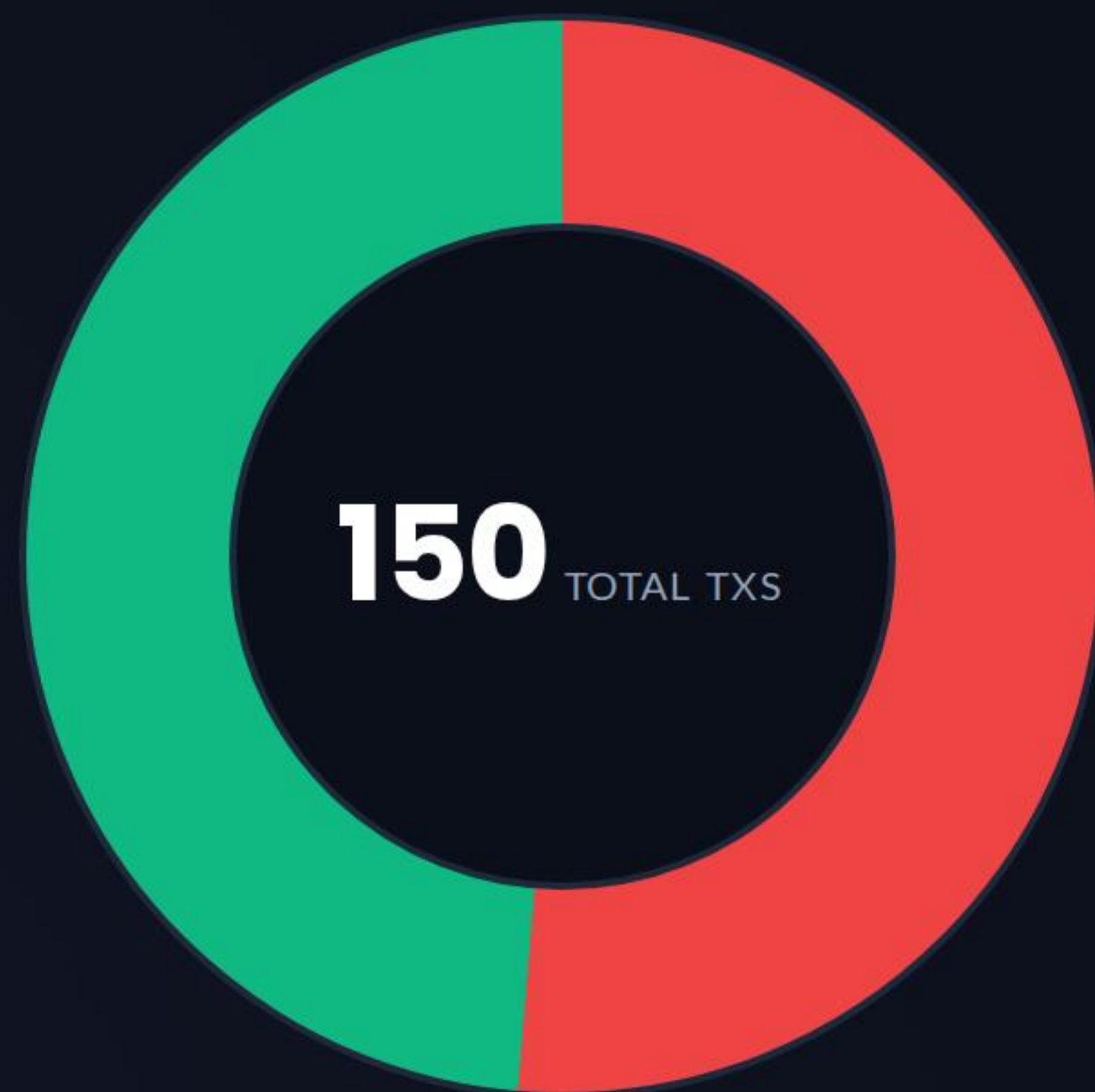
51.33% Failure Rate

77 Total failures out of 150 events

A critical systemic checkout threshold bottleneck has been identified. Immediate technical intervention is scheduled regarding bank API response thresholds and evening peak hour traffic balancing.

Success vs Failure Ledger Splits

GENERAL OPERATIONAL HEALTH



- Failed Transactions: 51.33%**
77 complete operational failures across merchant checkouts. Requires routing remedies.
- Successful Transactions: 48.67%**
73 successful transaction checkouts completely settled inside local server zones.

Frictional Failures by Category

CHECKOUT TIMEOUT DIAGNOSTICS

MERCHANT CATEGORY	REGISTERED STATUS	FAILED TRANSACTION OCCURRENCE	RISK METRIC
Entertainment	Critical	10 Failures	Immediate Action Required
Food	High Alert	9 Failures	Server Gateway Congestion
Shopping	High Alert	9 Failures	Server Gateway Congestion
Travel	Moderate	8 Failures	Standard Checkout Latency
Education	Moderate	8 Failures	Standard Checkout Latency
Healthcare	Moderate	8 Failures	Standard Checkout Latency

Spending Total Value by Category

ECOSYSTEM VOLUME INDEX



* Lowest Spends recorded inside **Utility Bills (₹28,974.21)** and **Fuel (₹31,289.68)**. Travel leads the ecosystem.

Ecosystem Transaction Frequency

VELOCITY DEMANDS

-  **Travel Bookings (18 Transactions)**
Represents highly planned, high-ticket conversions over digital checkout networks.
-  **Shopping (18 Transactions)**
Reflects frequent everyday lifestyle usage patterns across micro-merchant cores.
-  **Groceries (16 Transactions)**
Daily essential grocery checkouts supporting high routine pipeline volume metrics.
-  **Healthcare (16 Transactions)**
Urgent and essential utility payouts with high reliability expectations.
-  **Utility Recharges (15 Transactions)**
Data recharges and monthly telecommunications checkouts with stable cadence.
-  **Entertainment (15 Transactions)**
Leisure and booking transactions, heavily exposed to evening checkout failures.

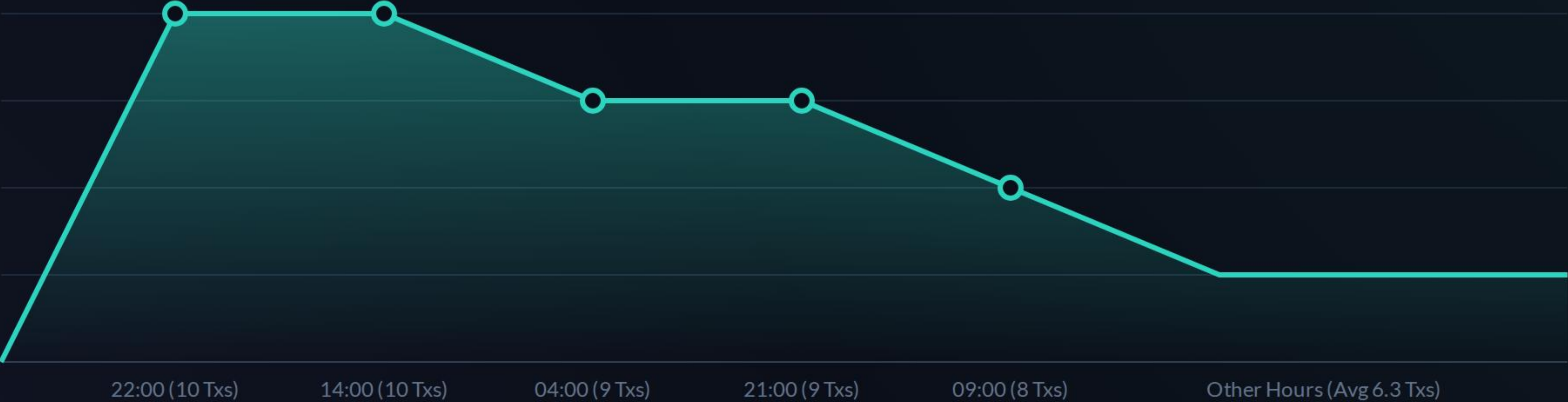
City-Wise Volume Leaderboard

GEOGRAPHIC ADOPTION MAPPING

Node Rank	City Name	City Tier Classification	Aggregated Transactions	Volume Density Rate
#01	Delhi	Tier-1 Metropolitan Core	27 Transactions	Highest Regional Peak
#02	Indore	Tier-2 Regional Growth Node	22 Transactions	Emerging Core Leader
#03	Jaipur	Tier-2 Regional Growth Node	16 Transactions	Stable Utility Baseline
#04	Chennai	Tier-1 Metropolitan Core	14 Transactions	Standard City Average
#05	Pune	Metropolitan Hub	13 Transactions	Standard City Average
#06	Kolkata	Metropolitan Hub	13 Transactions	Standard City Average

Peak Transaction Hours (24h)

TIME ZONE DENSITY METRICS



Diagnostics prove a strong bi-modal checkout distribution: Mid-Day (14:00) lunch payments & Late Evening (21:00 - 22:00) leisure checkouts.

Weekly Velocity: Weekdays vs Weekends

OPERATIONAL PATTERNS

Weekday Pipeline

102 Transactions

Daily Average: 20.4 Transactions/Day

Weekday volume accounts for 68% of the transaction dataset, primarily driven by routine merchant payments, daily grocery checkouts, and fuel purchases during transit slots.

Weekend Pipeline

48 Transactions

Daily Average: 24.0 Transactions/Day

Weekend daily volume densities experience a **+17.6% surge** compared to weekdays. This concentration points to recreational allocations (Dining, Entertainment, Travel Bookings).

High-Value User Concentration

ECOSYSTEM CONCENTRATION RISK

23.4%

VOLUME SHARE

Vulnerability Matrix

Our top 5 highest-spending consumer keys represent a combined transaction volume output of **₹96,606.86**, capturing nearly a quarter of all processed transactions inside the fact tables.

- User ID 7: ₹20,298.85 Spend (4.91% share)
- User ID 44: ₹19,966.97 Spend (4.83% share)
- User ID 25: ₹19,393.94 Spend (4.69% share)
- User ID 18: ₹18,671.51 Spend (4.52% share)
- User ID 20: ₹18,275.59 Spend (4.42% share)

Relational Database Architecture

DATABASE REFERENCE SCHEMA



upi_users (Dim)

Stores demographic metadata linking unique consumers.

- user_id [PK] (int)
- age (int)
- gender (text)
- city (text)
- city_tier (text)



upi_transactions (Fact)

Main ledger containing core transactional records.

- txn_id [PK] (int)
- user_id [FK] (int)
- txn_date (text)
- amount (double)
- merchant_category (text)
- status (text)



upi_merchant_categories

Dimension data classifying checkout endpoints.

- merchant_id [PK] (int)
- category [FK] (text)
- channel (text)
- avg_rating (double)

Business Strategy & Action Plan

STRATEGIC BLUEPRINT



Automated Gateway Failovers

Set up automatic real-time routing for Entertainment & Food checkouts during the evening peak traffic windows (21:00 - 23:00) to lower the high 51.33% transaction drop-offs.



VIP Customer Safeguards

Deploy dedicated backend pipeline channels and backup API routing tunnels for our top-tier active cohorts. This protects the core 23.4% transaction spend volume from gateway interruptions.



Targeted Growth Initiatives

Shift promotional budgets from oversaturated Tier-1 markets to high-adoption Tier-2 zones (such as Indore and Jaipur) where transaction velocities are showing outstanding natural growth.



Server Scaling Windows

Align system maintenance and bank partner gateway updates exactly around the off-peak slots (05:00 - 08:00) to keep high-impact business hours safe.

Thank You

UPI Transaction Performance Diagnostics analysis completed. Open for discussions, feedback, and system architecture recommendations.

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